

Crop Recommendation using Machine Learning and Explainable AI

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Abstract—Agriculture plays a significant role in economic development and food security, making crop selection an important factor for improving agricultural productivity. Choosing suitable crops based on soil characteristics and environmental conditions remains a challenge for farmers and may lead to reduced yield when traditional methods are used. In recent years, machine learning techniques have emerged as effective approaches for solving complex problems in agriculture.

This paper presents a crop recommendation system that utilizes important agricultural parameters including soil nutrients such as nitrogen (N), phosphorus (P), and potassium (K), along with environmental factors such as temperature, humidity, pH, and rainfall to predict the most suitable crop. Multiple machine learning algorithms including Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, Extra Trees, Gradient Boosting, and XGBoost were implemented and compared.

The performance of the models was evaluated using standard metrics including accuracy, precision, recall, and F1-score. Experimental results showed that the Extra Trees classifier achieved the highest prediction accuracy, while Random Forest was selected for detailed analysis because of its stability, consistency, and interpretability. In addition, SHAP (SHapley Additive Explanations) was employed to improve model transparency by understanding the contribution of individual features towards prediction outcomes.

The results demonstrate that integrating machine learning with explainable AI provides an accurate and interpretable solution for crop recommendation. The proposed system can assist farmers and agricultural stakeholders in making informed decisions, leading to improved productivity and efficient use of agricultural resources.

Index Terms—Machine Learning, Crop Recommendation, Random Forest, XGBoost, SHAP, Explainable AI, Agriculture, Classification

I. INTRODUCTION

Agriculture is one of the most important sectors contributing to economic development and food security, particularly in countries where a large portion of the population depends on farming for livelihood. Efficient agricultural practices are essential for improving crop productivity and ensuring sustainable utilization of available resources.

Crop growth and productivity depend on several important factors, including soil nutrients such as nitrogen (N), phosphorus (P), potassium (K), and environmental parameters including temperature, humidity, pH, and rainfall. These parameters collectively influence plant growth and agricultural output.

However, the relationships among these parameters are often complex and non-linear, making them difficult to analyze using traditional farming approaches.

Conventional crop selection methods generally rely on farmers' experience, historical knowledge, and manual assessment of agricultural conditions. Although such methods have been used for many years, they may not always provide optimal results because they often fail to consider multiple interacting parameters simultaneously. Therefore, there is a growing need for intelligent and data-driven systems capable of supporting accurate crop recommendation [1], [2].

In recent years, machine learning techniques have gained considerable attention in agricultural applications due to their ability to analyze large datasets and identify hidden patterns. Machine learning models can effectively capture complex relationships among different features and provide accurate predictions. Various algorithms including Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, and XGBoost have been successfully used in agricultural prediction tasks [3], [4].

In this work, a crop recommendation system is developed using multiple machine learning algorithms to predict the most suitable crop based on soil and environmental parameters. The dataset consists of important agricultural features such as nitrogen, phosphorus, potassium, temperature, humidity, pH, and rainfall. Different machine learning models are trained and evaluated using standard performance metrics including accuracy, precision, recall, and F1-score.

In addition to predictive performance, model interpretability plays an important role in practical applications. Therefore, SHAP (SHapley Additive Explanations) is employed to explain the predictions generated by the model and understand the contribution of each feature towards decision-making [5]. This enhances the transparency and reliability of the proposed system.

The main contribution of this paper is the development of an accurate, efficient, and interpretable crop recommendation system that combines the influence of both soil parameters and environmental parameters. Due to this integration, the proposed system achieves improved prediction performance and better decision-making capability. Furthermore, the parameters responsible for the overall performance of the system are investigated using SHAP-based explainable AI techniques,

which provide better transparency and understanding of the model predictions.

The remainder of this paper is organized as follows. Section II presents the background and literature review related to crop recommendation systems and agricultural prediction techniques. Section III discusses the basic concepts of the machine learning algorithms and explainable AI techniques used in this work. Section IV describes the proposed methodology, including data preprocessing, model development, evaluation, and system workflow. Section V presents the experimental results and analysis along with performance comparisons and model interpretation. Finally, Section VI concludes the paper and discusses future scope and possible improvements.

II. BASIC CONCEPTS

This section briefly discusses the machine learning algorithms and explainable AI techniques used in the proposed crop recommendation system. Different classification algorithms were implemented and compared to identify the most suitable model for crop prediction.

A. Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification tasks. It predicts the probability of a particular class using a logistic function. The algorithm is computationally efficient and commonly used as a baseline model for classification problems.

B. Decision Tree

Decision Tree is a supervised learning algorithm that uses a tree-like structure for decision-making. It divides the dataset into smaller subsets based on feature conditions and generates predictions through branching rules. Decision Trees are simple to understand and capable of handling non-linear relationships.

C. K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a distance-based machine learning algorithm that classifies data points based on the majority class among neighboring instances. The algorithm depends on selecting an appropriate value of K and is effective for pattern recognition tasks.

D. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised learning algorithm used for classification and regression problems. It works by identifying an optimal hyperplane that maximizes the separation between different classes. SVM can efficiently handle both linear and non-linear data.

E. Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting. In this work, both base Random Forest and tuned Random Forest models were implemented. Hyperparameter tuning was performed to optimize model performance and improve prediction accuracy.

F. Gradient Boosting

Gradient Boosting is an ensemble learning method that builds models sequentially by correcting errors made by previous models. Each new model attempts to minimize prediction errors and improve overall performance.

G. Extra Trees

Extra Trees (Extremely Randomized Trees) is an ensemble-based machine learning algorithm similar to Random Forest. However, it introduces additional randomness during tree construction by selecting random split points. This often improves generalization and reduces variance.

H. XGBoost

XGBoost (Extreme Gradient Boosting) is an advanced boosting algorithm designed for high performance and scalability. It improves prediction accuracy by sequentially reducing errors and is widely used in machine learning applications due to its efficiency.

I. Explainable AI

Explainable Artificial Intelligence (XAI) refers to techniques used to improve the transparency and interpretability of machine learning models. In this work, SHAP (SHapley Additive Explanations) is used to explain the predictions generated by the model. SHAP assigns contribution values to each feature and helps determine how different input parameters influence crop recommendation decisions.

III. METHODOLOGY

This section describes the complete methodology adopted for developing a crop recommendation system using machine learning techniques. The proposed approach follows a structured pipeline consisting of data collection, preprocessing, exploratory data analysis, model development, hyperparameter tuning, evaluation, and explainability. Each stage contributes to improving the overall performance, accuracy, and interpretability of the system.

A. Data Collection

The dataset used in this study consists of important agricultural and environmental parameters that influence crop growth and productivity. The input features include nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall, while the output represents the crop category. The dataset contains more than 2000 instances and is well-balanced across different crop classes, which helps in reducing bias and improving model generalization [1]. The availability of diverse and representative data plays a crucial role in building robust machine learning models.

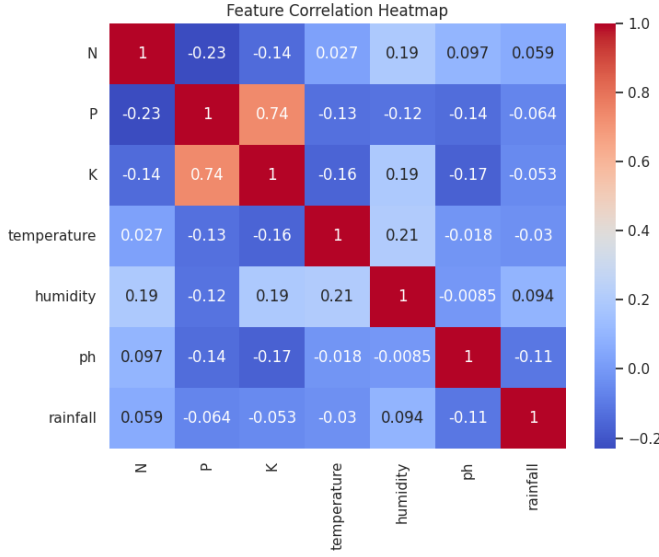


Fig. 1. Correlation heatmap showing relationships between soil nutrients and environmental features

B. Data Preprocessing

Data preprocessing is an essential step to prepare the dataset for model training. Initially, the dataset was examined for missing values, inconsistencies, and outliers. Since the dataset was clean, minimal preprocessing was required. Numerical features were standardized using scaling techniques to normalize the range of values, which significantly improves the performance of models such as KNN and SVM. The dataset was then split into training and testing sets to evaluate model performance on unseen data, ensuring that the model does not overfit.

To normalize the feature values and bring them to a common scale, min-max normalization was applied as follows:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

C. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the underlying patterns and relationships within the dataset. Histograms were used to analyze the distribution of features such as temperature, humidity, and rainfall, as shown in Fig. 1. A correlation heatmap was generated to examine the relationships between different variables and identify potential dependencies, as illustrated in Fig. 1. Additionally, the distribution of crop labels was analyzed to ensure class balance. This step provides valuable insights into feature importance and guides the model selection process.

D. Model Development

Multiple machine learning classification algorithms were implemented to predict the most suitable crop based on input features. The models include Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, Gradient Boosting, Extra Trees, and

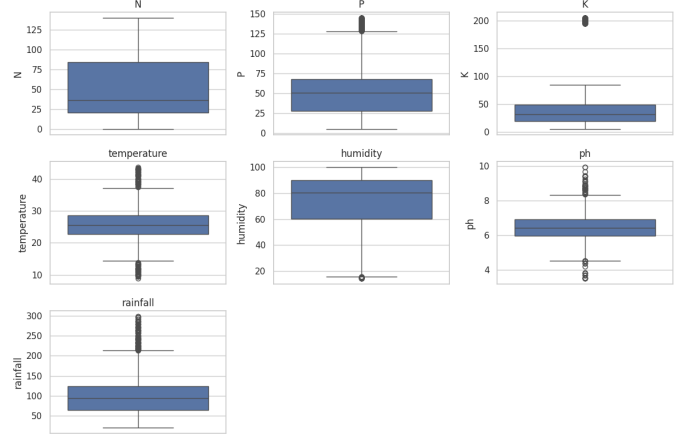


Fig. 2. Boxplot visualization of features showing distribution and presence of outliers in dataset

XGBoost. These models were selected due to their effectiveness in handling classification problems and their ability to capture both linear and non-linear relationships [3], [4]. Each model was trained on the processed dataset, and their performance was compared to determine the most effective approach.

E. Hyperparameter Tuning

To further enhance model performance, hyperparameter tuning was performed using techniques such as GridSearchCV and RandomizedSearchCV. Important parameters such as the number of estimators, maximum depth, and minimum samples were optimized to improve model accuracy and reduce overfitting. The tuned Random Forest model demonstrated superior performance due to its ability to aggregate multiple decision trees and minimize variance, making it highly effective for classification tasks.

F. Model Evaluation

The performance of all models was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive evaluation of the model's predictive capability. In addition, a confusion matrix was used to visualize the classification results and analyze the distribution of correct and incorrect predictions, as shown in Table 1. This evaluation process helped in selecting the best-performing model for deployment.

The performance of the models was evaluated using accuracy, defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

G. Explainable AI using SHAP

To enhance model interpretability, SHAP (SHapley Additive Explanations) was applied [5]. SHAP provides both global and local explanations by assigning importance values to each feature. The SHAP summary plot, shown in Fig. 5,

highlights the contribution of features such as nitrogen, phosphorus, potassium, and environmental factors in determining the predicted crop. This improves transparency and helps users understand the reasoning behind model predictions.

SHAP values are based on game theory and are computed as:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (3)$$

H. System Workflow

The overall system follows a structured workflow starting from data input and preprocessing, followed by exploratory data analysis and model training. The trained models are evaluated using various performance metrics, and the best-performing model is selected for detailed analysis and prediction. Multiple machine learning algorithms including Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, Gradient Boosting, Extra Trees, and XGBoost are implemented and compared. Although Extra Trees achieved the highest accuracy, Random Forest was selected for detailed analysis due to its stability, consistency, and interpretability. SHAP (SHapley Additive Explanations) is then applied to interpret the predictions and provide insights into feature importance. This integrated workflow ensures that the system is both accurate and interpretable, making it suitable for real-world agricultural applications. The complete workflow of the proposed system is illustrated in Fig. 3.

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IV. RESULTS AND ANALYSIS

A. Model Performance Comparison

The performance of various machine learning models was evaluated to determine the most suitable algorithm for crop recommendation. The models implemented include Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN),

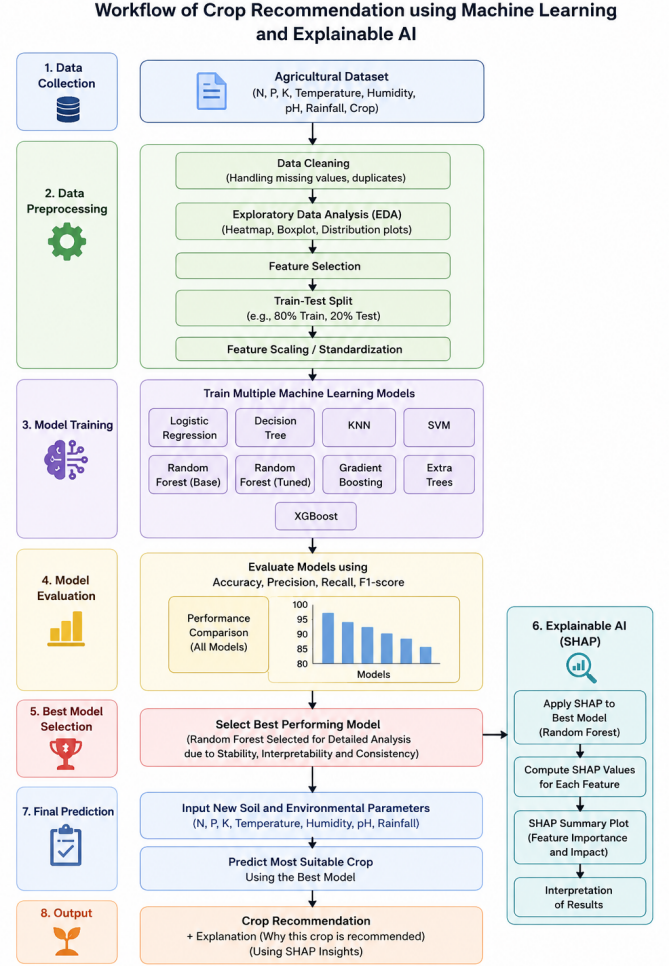


Fig. 3. Workflow of the proposed crop recommendation system using Machine Learning and Explainable AI

Support Vector Machine (SVM), Random Forest, Gradient Boosting, Extra Trees, and XGBoost. Each model was trained and tested using the same dataset under identical conditions to ensure a fair and unbiased comparison.

TABLE I
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Accuracy	Precision	Recall	F1 Score
Extra Trees	0.9955	0.9957	0.9955	0.9955
Random Forest (Tuned)	0.9932	0.9935	0.9932	0.9932
Random Forest (Base)	0.9932	0.9935	0.9932	0.9932
XGBoost	0.9909	0.9915	0.9909	0.9909
Gradient Boosting	0.9886	0.9897	0.9886	0.9887
SVM	0.9841	0.9856	0.9841	0.9840
Decision Tree	0.9841	0.9847	0.9841	0.9841
KNN	0.9795	0.9803	0.9795	0.9793
Logistic Regression	0.9727	0.9740	0.9727	0.9725

The comparative results of all models are presented in Table I. It can be clearly observed that ensemble-based models outperform traditional machine learning algorithms due to their ability to capture complex and non-linear relationships in

the data [3], [4]. Among all the models, the Extra Trees classifier achieved the highest accuracy and overall performance.

However, the Random Forest model demonstrated more stable and consistent results across all evaluation metrics. Although the Extra Trees classifier achieved slightly higher accuracy, the Random Forest model was selected for detailed analysis due to its stability, interpretability, and widespread use in practical applications. Additionally, Random Forest integrates effectively with SHAP-based explanation methods, making it more suitable for this study.

B. Evaluation Metrics

The models were evaluated using standard classification metrics including accuracy, precision, recall, and F1-score. Accuracy represents the overall correctness of predictions, while precision measures the proportion of correctly predicted positive observations. Recall indicates the ability of the model to identify all relevant instances, and F1-score provides a harmonic mean of precision and recall.

The evaluation results indicate that the ensemble models, particularly Extra Trees and Random Forest, achieved the highest values across all metrics. The balanced performance across precision and recall indicates that these models effectively minimize both false positives and false negatives, making them reliable for real-world agricultural applications.

C. Confusion Matrix Analysis

A confusion matrix was used to provide a detailed evaluation of the classification performance of the Random Forest model. The confusion matrix, shown in Fig. 4, illustrates the number of correct and incorrect predictions for each crop category.

Most of the values are concentrated along the diagonal, indicating correct predictions, while very few misclassifications are observed in off-diagonal elements. This confirms that the model has strong discriminative ability and can accurately classify different crop types based on the input features. The visualization also highlights the robustness of the model across multiple crop classes.

D. SHAP Analysis and Feature Importance

To enhance the interpretability of the model, SHAP (SHapley Additive Explanations) was applied [5]. SHAP provides a unified measure of feature importance by assigning contribution values to each feature based on its impact on the model's predictions.

The SHAP summary plot, shown in Fig. 5, highlights that soil nutrients such as nitrogen (N), phosphorus (P), and potassium (K), along with environmental parameters such as temperature, humidity, and rainfall, play a significant role in crop prediction. Features with higher SHAP values have a stronger influence on the prediction outcome.

Furthermore, SHAP analysis provides insights into how variations in feature values affect the prediction. For instance, higher nutrient levels and favorable environmental conditions positively contribute to the prediction of certain crops, while

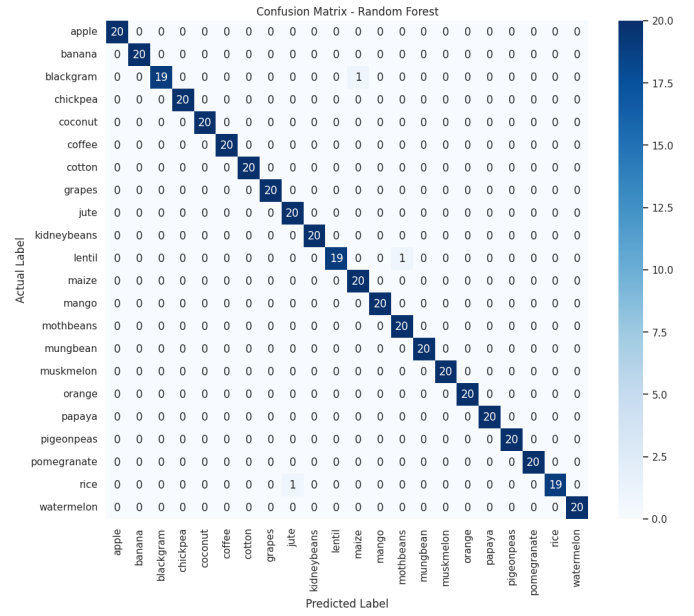


Fig. 4. Confusion matrix of Random Forest model showing classification performance across crop categories

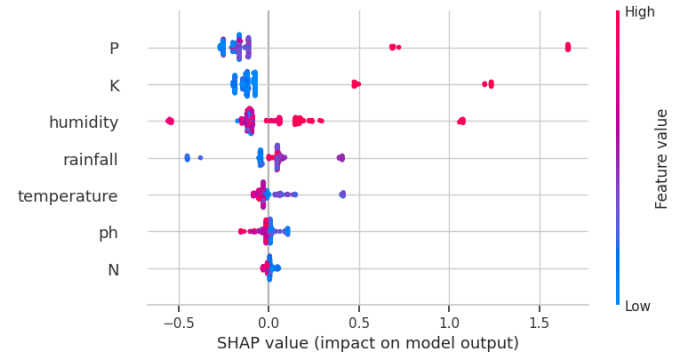


Fig. 5. SHAP summary plot showing feature importance and impact on model predictions

lower values may reduce their likelihood. This level of interpretability enhances trust and transparency in the model.

E. Comparative Analysis

A detailed comparative analysis of all models indicates that ensemble learning methods, particularly Extra Trees and Random Forest, outperform traditional models such as Logistic Regression and KNN. This is primarily because ensemble models combine multiple learners to improve prediction accuracy and reduce variance [3].

The results demonstrate that the use of ensemble techniques significantly enhances classification performance, making them more suitable for complex agricultural datasets where multiple features interact in a non-linear manner.

F. Overall Discussion

The overall findings of this study confirm that the proposed machine learning-based approach is effective for crop rec-

ommendation. The Extra Trees model achieved the highest accuracy, while the Random Forest model provided a better balance between performance, stability, and interpretability.

Additionally, the integration of SHAP improves model transparency and allows better understanding of the prediction process. The combination of machine learning and explainable AI techniques results in a system that is both accurate and interpretable, making it highly suitable for real-world agricultural applications where both performance and trust are essential.

V. CONCLUSION AND FUTURE SCOPE

A. Conclusion

In this study, a comprehensive machine learning-based crop recommendation system was developed to assist in selecting suitable crops based on soil nutrients and environmental conditions. The system utilizes key input features such as nitrogen (N), phosphorus (P), potassium (K), temperature, humidity, pH, and rainfall to predict the most appropriate crop category.

A wide range of machine learning classification algorithms were implemented, including Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, Gradient Boosting, Extra Trees, and XGBoost. These models were systematically trained and evaluated using standard performance metrics such as accuracy, precision, recall, and F1-score to ensure a comprehensive comparison.

The experimental results demonstrated that ensemble learning techniques, particularly the Random Forest classifier, achieved the highest performance among all models. The superior performance of Random Forest can be attributed to its ability to reduce overfitting and effectively capture complex, non-linear relationships between input features and crop categories. The tuned model consistently provided accurate predictions across different evaluation metrics, indicating its robustness and reliability.

Furthermore, the integration of SHAP (SHapley Additive Explanations) significantly enhanced the interpretability of the model. SHAP analysis provided valuable insights into the contribution of each feature towards the final prediction, highlighting the importance of soil nutrients and environmental factors in crop selection. This interpretability is crucial for building trust in machine learning systems, especially in real-world agricultural applications where decision transparency is important.

Overall, the proposed approach successfully demonstrates that combining machine learning with explainable AI techniques can lead to an efficient, accurate, and transparent crop recommendation system. This system has the potential to support farmers and agricultural stakeholders in making data-driven decisions, ultimately improving productivity and resource utilization.

B. Future Scope

Although the proposed system achieves high performance, there are several directions for future improvement and en-

hancement. One important extension is the integration of real-time data sources, such as live weather updates, soil moisture sensors, and satellite data, which can further improve the accuracy and applicability of the model.

Another potential improvement involves the use of advanced deep learning techniques, such as Artificial Neural Networks (ANNs) and hybrid models, to capture more complex patterns in agricultural data. These approaches may further enhance prediction performance, especially when dealing with larger and more diverse datasets.

The system can also be extended into a user-friendly application, such as a mobile or web-based platform, enabling farmers to easily access crop recommendations based on real-time inputs. This would increase the practical usability and impact of the system.

Additionally, expanding the dataset to include data from different geographical regions, soil types, and seasonal variations can improve the generalization capability of the model. Incorporating additional features such as fertilizer usage, irrigation patterns, and economic factors may further enhance prediction accuracy.

Finally, future work can explore the integration of Internet of Things (IoT) devices for automated data collection and real-time monitoring, enabling the development of a fully intelligent agricultural decision support system.

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